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Report for Raspberry Pi Face and Object Detection under Variable Lighting Conditions

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Abstract:

In this project, the system applying a real-time face and person detector with Raspberry Pi 4 and camera module is used to monitor a dynamic scene of four individuals and background objects within 20-second intervals in conditions of controlled lighting. The system needs to identify and label individuals and objects with pre-trained deep neural networks with real-time confidence scores and bounding boxes. The novelty of this work lies in the systematic testing of the system performance in the four different lighting conditions, including normal white (baseline) and red, green, and blue monochromatic light conditions. Detection log analysis of each scenario, however, shows that the generic person-detection class exhibits reasonable robustness (somewhere between 65 and 67 percent mean accuracy in normal and red light, falling to somewhere between 57 and 61 percent in the colored-light conditions), whereas face-recognition accuracy is much more sensitive to color casts, with the confidence decreasing as the lighting shifts to a colored (56 per cent) to a blue (52 to 54 per cent) one. Blue light is the most difficult one, and the total number of detections recorded is 48, as compared to 56-68 under other conditions. Through performance measurement in these lighting conditions, this project has been able to show how the performance of image sensors and Image Signal Processor (ISP) color balance can be influenced by real-world computer vision performance and provide recommendations to alleviate those situations including adaptive white balance and exposure correction.

Introduction:

System Requirements and Problem Statement

The challenge involves the development of computer vision system that will automatically identify and track a number of people and objects within a real-time video feed. The system should detect at least four individuals and one object in the scene at any time, display live bounding boxes with identity labels, confidence scores, generate a post-hoc plot of times of presence of each person; detect quality degradation errors due to ISP and sensor constraints.

Hardware and System Setup

The detection pipeline is executed on a Raspberry Pi 4 Model B featuring 4 GB of RAM and using Raspberry Pi Camera Module v2 (8 megapixel Sony IMX219 sensor) or other USB camera. IMX219 sensor has a 1/4-inch size, a fixed focus lens that is designed to see 50 cm down to infinity, and a pixel pitch of approximately 1.4 mm, able to detect the intensity of visible light as well as color temperature change. The Broadcom VideoCore IV ISP (Image Signal Processor) of the Raspberry Pi uses demosaicing, white balance and color correction in real-time, and tunable values of exposure, gain and saturation which directly affect downstream detection model performance.

The Pipeline of the Detection Model and Software

This system works with a pre-trained object detection model, but it is trained on the COCO dataset, and can detect 80-plus object types, such as person, refrigerator, remote, and other household objects. Face detection makes use of an independent face-recognition model that can identify up to four known faces (Sharon, Nikhil, Krishna, and Stimson in the study) but uses the generic classification of a person to identify an unrecognized one. The entire inference is executed on the Raspberry Pi processor; there is no support at all for any form of accelerated graphics processing and the real-time throughput is about 1-5 inferences/s depending on the complexity of the model it is running and the input resolution.

Reason behind Multi-Scenario Lighting Study

Majority of the computer vision systems are tested under ideal laboratory conditions (day light or controlled tungsten/LED arrays). In real-life applications, however, harsh color casts may be experienced because of monochromatic workplace lighting (factory floors using sodium vapor lamps or mercury vapor lamps), failure of the color correction in LED-based color correction systems, or colored lighting used in entertainment establishments. This project separates the effects of color information on generic object localization and identity specific face recognition by systematically testing the same detection pipeline on four separate lighting conditions (normal (balanced white), red (narrow-band red), green (narrow-band green), and blue (narrow-band blue)) and gives an understanding of sensor ISP tuning and model training needs to achieve a robust cross-lighting performance.

Methodology:

Experimental Design

Four video recordings of 20 seconds were taken consecutively, each having the same framing, background landscape, and positions of the actors but with various ambient levels of lighting. A detection log was created with each video recording using real-time inference of the video stream using the previously trained model, including timestamp, type of detection (face or generic object), known label, and score, and bounding box coordinates (top, right, bottom, left in pixel units).

Normal Lighting (Baseline)

The initial run was the baseline illumination, which was a standard white (CIE D65 or tungsten-corrected) under normal white (CIE D65 or tungsten-corrected) illumination and camera white balance was under automatic or daylight mode. This was a situation that was intended to be the best case performance and a benchmark against which all the future colored-lighting would be compared to. This detection log had 68 detections (between 0-19.93 seconds) of 4 faces (Sharon, Nikhil, Krishna, Stimson) and a large number of generic person detections and ad hoc object classes.

Red Lighting Scenario

A red light source (e.g., 650 nm LED array or gelled tungsten lamp) was placed to make the scene uniformly bright as a replacement of the usual lighting. The white balance of the camera was also deliberately not adjusted (or was switched to daylight /manual mode, in order to add as many red color cast as possible). This still allows realistic ISP operation: in field applications, automatic white balance can be turned off, incorrectly configured, or unable to respond quickly to sudden changes of light. The red-lighting location log had 56 frames of 0-19.87 seconds.

Green Lighting Scenario

The normal lighting was replaced with a light source of green color (e.g., 510 nm LED or gelled tungsten) and once more, camera white balance was maintained at the same level. Green-lighting detection log had 62 frames in 0-19.96 seconds.

Blue Lighting Scenario

A blue-colored source of light (e.g., 470 nm LED or gelled tungsten) was used, and the white balance was kept constant. The blue-lighting detection log recorded the least number of 48 frames in 0-19.94 seconds, the least of the four scenarios, and this means that there was a significant loss in detection confidence in this condition.

Information Gathering and Analysis

Reports of the detected objects were in the form of CSV files where the columns were: times (timestamp in seconds), type (face or object), label (identity or class name), confidence (0-100 scale), and bounding box coordinates. All downstream analysis was done in Python in pandas and numpy to compute per-identity and per-class statistics (mean, min, max confidence), detection counts and time distribution plot. The analysis using visual inspection of video frames and knowledge of scenes was done manually to identify false positives (spurious refrigerator, remote, cat, vase, and other objects that are not real) in the analysis.

ISP and Sensor Concerns

The Raspberry Pi Camera Module v2 ISP uses the Bayer filter and internal lookup tables to perform a fixed demosaicing process, auto white balance process as well as dynamic range compression. The white balance algorithm can also over-compensate or saturate under strong color casts resulting in clipping in the dominant color channel and loss of contrast in the other two channels . This channel imbalance has a direct impact on neural network feature extraction: networks trained on balanced RGB data are unable to work when fed with extreme changes of colour or saturated channels. It can also happen that the ISP can use very strong noise reduction or temporal smoothing, which blurs fine details that are important in face recognition, at low light or under narrow-band lighting.

Results:

The data demonstrates the evident tendency, deviation of the lighting to the monochromatic color instead of the balanced white decreases the overall number of detections and decreases the average scores of the confidence. The largest decrease is seen when using blue lighting, in which overall detections are reduced to 48 (29% lower than the normal baseline) and average object confidence is 56.92, the lowest of all. This trend is indicative of the camera sensor (poor spectral response in blue wavelengths at blue pixel of the Bayer filter) and the neural network (responds to color-shifted images in general) being sensitive to color-shifted inputs.

Metric	Normal	Red	Green	Blue
Total Detections	68	56	62	48
Mean Face Confidence (%)	56.03	52.07	54.22	52.34
Mean Object Confidence (%)	67.19	65.19	60.78	56.92
Named Faces Detected	4	4	3	2
Person-Class Detections	37	43	44	15

Normal Lighting Situation

The system is the most reliable when used under normal (balanced white) illumination. Sharon is detected 0.00-3.75 s, Nikhil 6.51-11.68 s, Krishna 8.01-11.68 s and Stimson 15.69-18.67 s. The face-confidence scores are about 56, with an individual score as low as 51.49 (Sharon) and high as 62.19 (Krishna). High consistency is observed in the generic person class 37 times (mean 67.19%), meaning that the coarse human silhouette is consistently found with typical light. There are 10 entries of false positives (refrigerator, remote) which are presumably due to background clutter confusing the object detector, but which should not be considered as evidence of model failure as they are low-confidence (50-69) and do not affect the actual face-recognition task.

Red Lighting Scenario: Early Degradation

Face identifications decrease to 13 under red lighting in the four identities. Mean face confidence goes down to 52.07 percent- an absolute change of 7 percent of baseline. Per-identity analysis demonstrates that Nikhil has a reasonable performance (~55.63%), whereas Sharon and Stimson end up with less than 50, indicating that red illumination is especially damaging to the face with a cooler skin color or with a face utilizing subtle features of the green and blue channels to be recognized. On the other hand, the detections of person in class are also strong with 43 instances and mean confidence of 65.19 which implies that silhouette-based detection is not much sensitive to the red color cast. It is important to note this asymmetry: the ISP white balance can enhance the noises in the green and blue channel when the red light is captured, thus ruining fine facial features, but not gross shape information.

Green Lighting Case

The green light yields 62 combined detections, than the red one. Face detection revives to some extent, 11 named-face entries are compared to 13 in red. The system however generates many spurious object labels: cat, cell phone, vase, toilet, handbag, and refrigerator- some of which could be real (e.g., where an actor was holding a phone), but others are obvious hallucinations of grey background texture under a green light. Mean object-class confidence is 60.78% and it is the first time that average object-class confidence is less than 61. This indicates that the green light that is not very perceptive to the human skin tone and tends to saturate the ISP of the green channel is especially disruptive to object classification. Faces are still at 54.22% mean confidence close to the red but still lower than the baseline.

Blue Lighting Scenario

The worst performance failure comes with blue light. The total number of detections is recorded as only 48 (a 29% decrease compared to the baseline), and a total of only two identities are identified: namely, Nikhil (15 times) and Krishna (1 time). Mean face confidence is 52.34 which is similar to red and only a small number of detections are required to reach that confidence level and it means that most scenes in the blue-lighting video do not cause enough confidence to be included in the log. Detections of objects are at their lowest point of 15 "person" instances and a mean of confidence of 56.92. Simultaneously, the system has high numbers of false-positive labels: 13, 2, and 2 times of the labels of tv, vase, and cat, respectively. The blue case proposes a failure mode in which the blue channel of the Bayer filter is either clipped (because the ISP was saturating its ISP gain with the weak blue light) or very noisy (because the ISP was attempting to scale up a weak blue signal), which causes unreliable extraction of RGB features and inference of the model.

Why Blue is Worse

Red and green wavelengths are associated with high levels of reflectance in human skin, whereas blue wavelengths are associated with lower levels of reflectance in human skin (10-20% reflectance range across all skin types). In blue monochromatic lighting, the camera gets low relative signal in the red and green channels (reflecting darker skin, mostly) and medium signal in the blue channel. It is then followed by the Bayer demosaicing algorithm which tries to reconstruct three channels using undersampled data and the ISP uses aggressive gain to compensate the low overall light-an operation which encourages noise in weak channels. On the other hand, red and green monochromatic scenes are at least very strong signal in at least one of the Bayer channels and can be partially reconstructed in color. And the deep neural networks that are trained on the COCO dataset also have a higher portion of training examples of objects in warm (red/yellow) illumination compared to cool (blue) illumination, hence generalize well to reddish-cast inputs.

Continuity of Detection

In all the cases, Nikhil is most reliably identified (6 times in normal, 6 in red, 4 in green, 15 in blue) and Sharon and Krishna are more varied. This is probably a combination of pose, facial geometry, distance to camera and the ISP processing, but not the bias of training data. The order of entry and exit times is similar in the normal and red cases (Sharon - Nikhil - Krishna and Nikhil together - Stimson) and this indicates that the video content is the same across the runs and the system is reacting to actual changes in light and not to changes in the scenery and content.

Mitigation Measures and Future Advancements:

The results indicate that a number of measures can be applied to enhance performance during colored lighting:

- **Adaptive White Balance**: Buttons Enable or tune the Raspberry Pi ISP white balance to be more straightened in its solution of color casts. This would enhance the red and green channels under the blue light, somewhat regaining balanced RGB appearance.
- **Channel-Specific Gain Control**: A dynamic per-channel exposure control would allow weak channels (when the red and blue colors are both under blue light) to remain in the noise floor.
- **Training the models**: Fine-tune the object and face-detection models on augmented data with colored illumination, either on synthetic data or real data, either with domain adaptation or color normalization techniques.
- **Normalization Preprocessing**: Convert color-space transformations - prior to feeding frames to the neural network, histogram apply histogram equalization or other color-space transformations (e.g. HSV or LAB) to expel the effects of global color variations.
- **Ensemble Inference**: Operate many models (one per possible lighting condition) simultaneously and combine their results, at a cost of trade-off in the cost of computation.

Ethical Implications

Although computer vision surveillance and person recognition involves well-reported privacy and bias issues, another ethical aspect revealed by this project is the unfair performance in varying light intensities, since such a project may be skewed towards the better-recognition of people of darker complexion, or those who are generally represented in colored light in training examples. As illustrated by the example that a model trained on footage of people in warm (red/yellow) lighting (typical of many datasets) will produce low accuracy on the same people in cool (blue/green) lighting, the model will have a lighting-based recognition bias. Implementation of these systems in actual surveillance or access-control applications without being aware of these restrictions may result in false negatives (not identifying authorized people) or false positives (identifying the wrong people), and that may have implications on fairness and due process. The steps that organizations implementing computer vision systems ought to take include; assessing their performance over a variety of lighting conditions prior to deployment, making known limitations public to users, recording all their detections and confidence values to facilitate auditability, and availing human-in-the-loop review on high-stakes decisions.

Conclusion:

The project shows that it is possible and quite acceptable (with regular balanced light, 56% mean face confidence, 67% mean object confidence) to perform real-time face and object detection on a resource-constrained platform (Raspberry Pi). Nevertheless, dirty lighting in monochromatic colored affect performance significantly and asymmetrically: red lighting causes an absolute decrease in face and object confidence by 7 and 7 percent, respectively; green lighting causes an absolute decrease in face and object confidence as well as spurious object labels; and blue lighting causes an absolute decrease in total detections and gross misclassification. This is because of the interaction of spectral sensitivity of the Bayer color filter, white balance and gain-control algorithms of the ISP and those features learned by the neural network that are not consistent across out-of-distribution color shifts.

The insights gained through this work in practice are:

- Presence is less important than identity, which is dominated by lighting. Generic person detections are approximately 10% more resistant to color casts than face-recognition detections, indicating that the applications which just need counting or presence detection are more light-resistant than those that need identity.
- Blue is essentially more difficult than either red or green. This is probably a result of the blue-channel sensitivity of the Bayer filter, as well as biases in the training-data. Deployments with blue lighting need to be more focused on white-balance correction or retraining models.
- Tuning of ISP is of equal importance as model choice. This neural network when trained on well-balanced RGB images (through good ISP settings) achieves tremendously better results using color-shifted inputs, an observation that may indicate that meticulously calibration of both sensors and ISP is required before deployment of a given model.
- Color lighting is an indirect measure of greater robustness issues. The results of this research apply to other out-of-distribution cases (snow, fog, extreme shadows, etc.) and thus suggest that computer vision practitioners need to test systems in a variety of environmental conditions and not only in optimal laboratory conditions.

The next round of research must be on automated adaptive white balance, multi-model ensembles, and color-agnostic features (e.g. edge-based or depth-based detection) as a way of enhancing robustness in real-world applications.